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JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

[A GOVERNMENT AUTONOMOUS COLLEGE]

JGEC/B.TECH/CSE/IT/BS-M201A/ 2024-25

2025

Mathematics-IIA

Full Marks: 70 Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions $5\times2=10$

1. Show that $\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}(X)$. 2
2. Find the probability that in 8 throws of a die, the number 1, 3, 5 turn up 2, 3, 3 times respectively. 2
3. If X and Y are independent random variables, then show that they are also uncorrelated. 2
4. If $\mathrm{Var}(X)=9$, $\mathrm{Var}(Y)=4$ and $\mathrm{Var}(X-Y)=\mathrm{Var}(X)$, then find the correlation coefficient between X and Y. 2
5. Suppose that the probability of an item being defective, in a mass production process of that item is 0.01. If 20 items are selected at random, then what is the probability that exactly 2 will be defective. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *five* questions $12\times5=60$

6. (i) Two person agree to play a game by drawing balls by turn from a box containing 4 white and 6 black balls. He who draws the first white ball wins. Find the probability that the man who starts the game loses the game. 4

(ii) Three urns contain respectively 1 white and 2 black balls, 2 white and 1 black balls, 2 white and 2 black balls. One ball is transferred from the first to the second urn, then one ball is transferred from the second to the third urn, finally one ball is drawn from the third urn. Find the probability that the ball drawn is white. 4

(iii) Show that the probability of obtaining six at least once in 4 throws with a die is slightly greater than 0.5. 4

7. (i) A discrete random variable X has the following probability mass function

X=x	1	2	3	4	5	6	7
P(X=x)	k	2k	2k	3k	k^2	$2k^2$	$7k^2$+k

(a) Determine the constant k 1+
(b) Evaluate $P(3<X|X\le6)$ 1+
(c) Find the minimum value of x so that $P(X\le x)>\frac{1}{2}$ 1+
(d) Obtain the distribution function $F(x)$ 2+
(e) Find mean of X. 1+
(ii) (a) Determine the value of c such that $f(x)$ defined by 2+
$f(x)=\begin{cases} c(x-1)(2-x), & 1<x<2 \\ 0, & \text{elsewhere}\end{cases}$ is a probability density function. 2+
(b) Find the corresponding probability distribution function. 2
(c) Also find $P(5<X<6)$.

8. (i) The length of bolts produced by a machine is normally distributed with parameters $m=4$ and $\sigma=0.5$. A bolt is defective if its length does not lie in the interval (3.8,4.3). Find the percentage of defective bolts produced by the machine.$\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.6}e^{-\frac{t^2}{2}}dt=0.7257, \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.4}e^{-\frac{t^2}{2}}dt=0.6554\right]$ 4

(ii) Show by Chebycheff's inequality that in 2000 throws with a coin the probability that the number of heads lies between 900 and 1100 is at least $\frac{19}{20}$ 4

(iii) A secretary writes four letters and the corresponding address on envelopes. If he inserts the letters in the envelopes at random irrespective of adress, then calculate the probability that the letters are wrongly placed. 4

9. (i) Calculate mean, median and find the approximate value of the mode from the following frequency distribution: 6

Height :	60-63	64-67	68-71	72-75	76-79	80 -83
No of students:	8	3	18	6	16	8

(ii) If the pdf of a continuous random variable X is given by 3+3

$$f(x) = \begin{cases} xe^{-x^2} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

then find the mean and variance of the distribution. 6

10. (i) Marks obtained by 10 students in Physics (X) and Mathematics (Y) are given in the following table :

X: 48 33 40 9 16 16 65 24 16 57

Y: 13 13 24 6 15 4 20 9 6 19

Find the Rank correlation of the two series of marks. 3+3

(ii) Following is a grouped frequency distribution of expenditure of 1000 families. The mean and median of the distribution are both 87.5. Find the missing frequency:

Expenditure	40-50	60-79	80-99	100-119	120-139
No .of Family	50	--	500	--	50

6

11. (i) Find the skewness and kurtosis of the following frequency distribution:

X	112	117	122	127	132	137	142
f	5	15	20	35	10	10	5

(ii) The relationship between travel expenses (y) and duration of travel (x) is found to be linear. A summary of dates for 102 pairs is given below: 3+3

$$\sum x = 510, \sum y = 7140, \sum x^2 = 4150, \sum xy = 54900, \sum y^2 = 740200.$$

(a) Find the two regression coefficients

(b) Find the two regression equations. 6

12. (i) Fit a second degree parabola $y = a + bx + cx^2$ to the following data by method of least square:

X	1	2	3	4	5
Y	2.18	2.44	2.78	3.25	3.83

(ii) In a sample of 500 students drawn from the college students in West Bengal it is observed that the number of students reading in 1st Year, 2nd Year, 3rd Year and 4th year are 200, 150, 100 and 50 respectively. Will it support the hypothesis that in West Bengal the number of students reading in 1st Year, 2nd Year, 3rd Year and 4th year are in proportion 3:2:2:1? . Test at 5%level of significance. Given that 5% value of chi square for 4 d.o.f is 7.815 and 7.416 respectively. 6

13. (i) For two variables x and y, the two regression lines are x+4y+3=0 and 4x +9y+5=0. Identify 3+ 1+ 1+

which one is of y on x. Find means of x and y. Find the correlation coefficient between x and y. 1

Estimate the value of x when y= 1.5.

(ii) Find the variance and standard deviation of the following frequency distribution: 6

X	30-35	41-45	46-50	51-55	56-60	61-65	66-70
f	14	26	40	33	50	37	25

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[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/CSE/ECE/IT/BS-PH201/2024-25

PHYSICS

Times: 3 Hours

Full Marks: 70

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

$5 \times 2 = 10$

Answer *all* questions

1. Two particles have position vectors given by $\vec{r_1} = 2t\hat{\imath} - t^2\hat{\jmath} + (3t^2 - 4t)\hat{k}$ and $\vec{r_2} = (5t^2 - 12t + 4)\hat{\imath} + t^3\hat{\jmath} - 3t\hat{k}$. Find the relative velocity of the second particle with respect to the first at the instant $t = 2s$. 2

2. Prove that the logarithmic decrement is the time required for the maximum amplitude during an oscillation to reduce to *1/e* of this value. 2

3. An electron microscope uses 1.25 KeV electrons. Can you comment on its resolving power? 2

4. A sheet of cellophane is a half-wave plate for light of $\lambda = 4 \times 10^{-5}$ cm. Assuming that there is negligible variation in indices of refraction with wavelength how the sheet would behave with respect to light of wavelength 8×10^{-5} cm? 2

5. Show that the displacement current density leads the conduction current density by $\frac{\pi}{2}$. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

$4 \times 15 = 60$

Answer any *four* questions

6. a) A charge particle $q = -10\ \mu C$ is carried along the shortest path from (0, 0) to (2, 0) and (2, 0) to (2, 2) and then (2, 2) to (0, 0) in an electric field $\vec{E} = (x + 2y)\hat{\imath} + 2x\hat{\jmath}$. Find the work done by an external agent in (i) each path and (ii) the round trip. 4

   b) Potential for a conservative force field is given by $\varphi = 3x^2y + xz^2 + 4$. Find the direction of the force at a point (1,0,1). 2

   c) Show that $r^n\vec{r}$ is irrotational vector for any value of n but is solenoidal if $n = -3$, $\vec{r}$ being the position vector of a point. 5

   d) A particle of mass m is initially at rest at the origin. It is subjected to a force and starts moving along x-axis. Its kinetic energy K changes with time as $dK/dt = \gamma t$, where γ is a positive constant of appropriate dimensions. Show that the force applied on the particle is constant and the speed of the particle is proportional to time. 4

7. a) A single slit is illuminated by light composed of two wavelengths. One observed that due to diffraction, the first minima obtained for one wavelength coincide with the second diffraction minima of another wavelength. Calculate the ratio between the wavelengths of the light. 3

   b) The electric field E of a plane wave in air is given by $\vec{E} = \hat{\imath}4 \times 10^{-6}cos(10^7\pi t - kz) + \hat{\jmath}4 \times 10^{-6}sin(10^7\pi t - kz)$ V/m. Find the value of k, the magnetic field and Poynting vector. 4

   c) Given that $\vec{B} = \vec{\nabla} \times \vec{A}$ and $\vec{\nabla}.\vec{A} = 0$. Then show that $\nabla^2\vec{A} = -\mu_0\vec{J}$. 3

   d) Using Maxwell's equation derive wave equation for electromagnetic waves in a medium having finite permeability and permittivity but no conductivity. 5

8. a) Show that the de-Broglie wave group associated with a moving particle travels with the same velocity as that of the particle. 4
b) What is Compton effect? Derive an expression for Compton shift. 5
c) Find the energy of the neutron in unit of eV whose de Broglie wavelength is 1 $\mathring{A}$. 3
d) In an experiment tungsten cathode which has threshold 2300 $\mathring{A}$ is irradiated by UV light of wavelength 1800 $\mathring{A}$. Calculate (i) maximum energy of the photo electron (ii) work function for tungsten. 3

9. a) If the pass axis of a polarizer is rotated very slowly, plot the variation in intensity of the emergent beam for the incident linearly, circularly and unpolarized beam of light. 3
b) Write a short note on half-wave plate. 4
c) Comment on the state of polarization for the following light wave: $\vec{E} = \hat{j}\, A\cos(kx - \omega t) + \hat{k}\, 2A\cos(kx - \omega t + \pi/4)$ 3
d) Derive Brewster's law. Hence calculate the polarizing angle for glass pate ($\mu_g = 1.5$) immersed in water ($\mu_w = 1.33$). 5

10. a) A sugar ball of radius R dissolves in excess water in time T. Find out the required time to dissolve the sugar ball having radius 3R. 3
b) A series LCR circuit is subjected to a sinusoidal emf $V = V_0 \cos \omega t$. Find the amplitude of current. Discuss the phenomenon of resonance in such circuit. 4
c) In a photoelectric experiment a parallel beam of monochromatic light with power of 200 W is incident on a perfectly absorbing cathode of work function 6.25 eV. The frequency of light is just above the threshold frequency so that the photoelectrons are emitted with negligible kinetic energy. Assume that the photoelectron emission efficiency is 100%. A potential difference of 500 V is applied between the cathode and the anode. All the emitted electrons are incident normally on the anode and absorbed. Find the force experienced by the anode due to the impact of the electrons. 4
d) Find the resultant of N simple harmonic motion all having the same amplitude and with their phases increasing in arithmetic progression 4

11. a) An atom of oxygen on being polarized produces a dipole moment of 5×10^{-23} Cm. If the distance of the center of negative charge cloud from nucleus be 4×10^{-15} cm, calculate the polarizability of the oxygen atom. 4
b) Derive Clausius-Mossotti equation. 5
c) Write a short note on dielectric loss. 3
d) A capacitor with two parallel plates of dimension 1 cm × 2 cm receives 2.25 V potential difference between electrodes. (i) How far apart must the plates be to produce a charge density of 10^{-7} C/m^2? (ii) How many electrons accumulate on the negative plate? [No dielectric is there] 3

Useful constants

1. Speed of light: 3×10^8 m/s
2. Planck's constant: 6.626×10^{-34} J.s
3. Electron charge magnitude: 1.6×10^{-19} C
4. Electron rest mass: 9.109×10^{-31} kg
5. Proton rest mass: 1.672×10^{-27} kg
6. Neutron rest mass: 1.675×10^{-27} kg
7. Free space permittivity: 8.8542×10^{-12} C^2/N-m^2
8. Free space permeability: $4\pi \times 10^{-7}$ N-sec^2/C^2

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JGEC/B.TECH/ CSE/IT/ECE /ES-CS201/2024-25

2025

PROGRAMMING FOR PROBLEM SOLVING

Full Marks: 70 Times: 3 Hours

The figures in the margin indicate full marks.

Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A

[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions 5x2=10

1. Convert $(2AB)_{16} = (?)_{10}$

2. What is the output of the following code?

```
char name[ ]= "Computer Science and Engineering";
printf("%d",strlen(name));
```

3.
```
#define JGEC(x)(x*10)
void main( )
{
int a=3, b;
   b= JGEC(a + 2);
printf("\n%d",b);
}
```
What will be the output?

4. Which operator can be used to access structure data members if the structure data is accessed using structure to pointer variable?

5. What is the associativity of the arithmetic operators?

GROUP-B

[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions 4x15=60

6. i) Write down the difference between break and continue with suitable example. 5 + (3+2) +5

ii) Write down the advantages and disadvantages of "switch" over "if-else"? What are the properties of an array?

iii) Let a 2 D array is declared as int a[2][3]. What is the total memory size allocated by this array and maximum how many elements can be stored in this array? If the base address is 1000, compute the address of a[1][1] considering both row major and column major manner.
[Size of an integer variable 4 bytes].

7. i) Write a Program in C to find & print the Sum of all the Numbers Divisible by 7 within a given range. 5+ 5 +5

ii) Write a C program to print the lower triangular of a given square matrix.

iii) Write a program to determine and print the sum of the following harmonic series for a given value of n: 1+1/2+1/3+.....+1/n. [8+7 marks].

8. i) Explain the different types of loops in C with syntax. 5 +5 +5

ii) Write a C program to compute the sum of the two input values. If the two values are the same, then return triples their sum.

iii) You have recharged your mobile for the given (user given) amount. The service provider offers the following facilities

Within same service provider	10 paise per call
With other service provider	30paise per call
STD calls(with other or within service provider)	60 paise per call
SMS	20 paise per call

Write a program to read the number of calls and sms made by you and calculate the amount deducted for your calls and sms. Also display the balance

9. i). Explain call by value and call by address method with an example 5

ii) Write a program to find the number of occurrences of a given character in your name. +5

iii) Write a suitable code in C language to find the Fibonacci series up to Nth term using recursive function. +5

10. i) Admission to a professional course is subject to the following conditions: 7+

a. marks in maths>= 60. 3+

b. marks in physics >= 50. 5

c. marks in chemistry >= 40.

d. total in all three subjects >= 200.

Given the marks in three subjects, write a C program to process the application to find whether a candidate is eligible or not.

ii) What is ternary operator? Explain with suitable example.

iii) Distinguish between an array of structures and an array within a structure. Give an example each.

11. i) Write the output of the following code and explain the output 5+

```
        int main(){
        int i=xy(20);
        printf("%d",--i);}
    int xy(int j)
      {
       return(j++);}
```

5+ (3+2)

ii) Write a program in C to calculate the power of any number using recursion

iii) What are the different modes of file opening? Write the function of strcat() and strcpy() function.

12. Write short notes on any three 5+ 5+ 5

i) Pointer Arithmetic

ii) Storage Classes

iii) Operators in C language

iv) Formal arguments and actual arguments

v) Structure pointer

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ENGLISH

l Marks: 70 — Time: 3 Hours

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Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

nswer *all* questions — 5x2=10

Correct the error in the sentences.

1. The captain along with his team are practising very hard for the forthcoming matches. 2
2. Supposing if it rains what shall we do? 2
3. Everyone knows that the leopard is the faster of all animals. 2
4. Ram was senior than Sam in college. 2
5. It was him who came running into the classroom. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions — 4x15 = 60

6. i) Communication is a process of sending and receiving information. Explain the communication process in the light of this statement. 10

ii) Fill in the blanks with appropriate prepositions 5

a) When you believe injustice is being done stand ________ the authority.
b) She puts _______ two hours on her English studies every day.
c) Your strategy ultimately paid _____ and we are reaping the benefits.
d) She excels _____ mathematics.
e) Please shut the door _______ you.

7. i) Wanted a Plant Manager (operations)at our new factory in Gaziabad, UP. Engineering Graduates with minimum 5 years' experience in manufacturing industries can apply. Salary negotiable. Apply with particulars to Box 650, The Hindu, Chennai-600004. 10

ii) One-word Substitutions: 5

a) A famous person:
b) Divide into two parts:
c) Extreme fear of being in a small enclosed place:.
d) One who runs own business:
e) Belonging to the same period:

8. i) Realizing the need of packing services in Faridabad, Elite Professional Packers have recently started their services in the city. You, as the publicity manager of this company have the onus of promoting this service. Draft a sales letter to be sent to the prospective users. 10

ii) Use the correct form of tense in the blanks 5

Marie had probably caught my odd stare at the table and so _______ (move) over and blew the candle out. She then ________ (glance) at the window and ________ (go) over to open them. Father was _________ (sit) on the bed and said in a cheerful voice, "I ______ (like) the country air. It suits me well."

9. i)Imagine yourself to be the instructor of a course in which 75 students have registered. Draft an email to be sent to all these students asking them to select a topic of their choice and prepare for a professional presentation. 10

ii)Transform the sentences according to the instruction given: 5

a) Horses are bigger than donkeys. (begin: Donkeys aren't...)

b) Few people know me. (begin: Not...)
c) Children drink more milk than adults. (begin: adults don't...)
d) There is little water. I can't have a bath. (begin: there isn't enough...)
e) Reading isn't as interesting as travelling. (begin: travelling is...)

10. i) Write an essay on the future of engineering education.
ii) Make sentences to point out the difference between the following pair of words:
a) which/witch b) vein/vain c) yoke/yolk d) weak/week e) rain/reign

11. i) Write a note on the various levels of communication.
ii) Choose the correct synonym of the underlined words from the list given below:
(brushed, grasped, disagreed, steered, soil)
a) I was digging in the dirt when my mother called me for lunch.
b) Mike groomed his dog.
c) Noah argued with his brothers.
d) Matt grabbed her hand
e) Rob drove his truck to school.